monitoring contributes directly to wildfire prevention and grid reliability, which are major public safety concerns in the region.

These programs illustrate how UAVs support a shift toward data-driven infrastructure resilience. By improving inspection frequency, accuracy, and safety, UAVs enable municipalities and utility providers to transition from reactive repairs to predictive maintenance strategies. Despite challenges such as regulatory airspace limitations and the need for skilled data analysts, the success of UAV-based inspection initiatives in New York and California demonstrates the long-term value of integrating drones into urban asset management and public safety infrastructure.

8.10.5 Case study: India – Smart City Mission UAV integration

India’s Smart City Mission has encouraged the adoption of UAV technologies to support urban governance, planning, and public service delivery in rapidly growing metropolitan areas. Cities such as Hyderabad, Varanasi, Surat, Bhopal, and Bengaluru have integrated drones into civic workflows to enhance monitoring, mapping, and decision-making capabilities. One of the major applications is high-resolution urban mapping, where UAVs generate 3D city models and geographic information system (GIS) layers to aid in land-use planning, infrastructure expansion, and redevelopment projects.

These maps allow municipal authorities to identify encroachments, assess informal settlements, plan road widening projects, and optimize public space usage with greater accuracy than traditional survey methods (Raman, et al. 2021).

UAVs have also become valuable tools in solid waste management and sanitation monitoring, particularly in cities such as Surat and Indore. Drones are used to track waste collection vehicle routes, identify illegal dumping hotspots, and monitor landfill capacity. By overlaying drone data